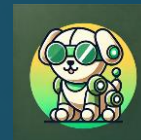


Guide Dog

Shared vision pipeline for the euROBIN VIP demo



Motivation

Robot use case

- **User: “We have a camera on our robotic system but no object pose estimation”**
- **User sends image to obtain full 6D pose for all visible objects**
- **Registration for user requests from outside the institute and authentication**

Prototyping use case

- **User: “I have a camera and just need quickly the 6D pose of this object for some tests”**
- **Potentially requested from within the institute**

Guide Dog to the rescue!



Guide Dog: A vibe-coded PoC

Setup camera

Please enter your camera parameters:

camera_name: robot_camera
robot_camera

fx: 620.0 fy: 620.0
620.000 - + 620.000 - +

ppu: 640 ppv: 340
640 - + 340 - +

width: 1280 height: 640
1280 - + 640 - +

Set camera parameters

Image upload

Upload an image

Drag and drop file here
Limit 200MB per file • JPG, PNG, JPEG

Browse files

image_000080.png
1.2MB

Delete image

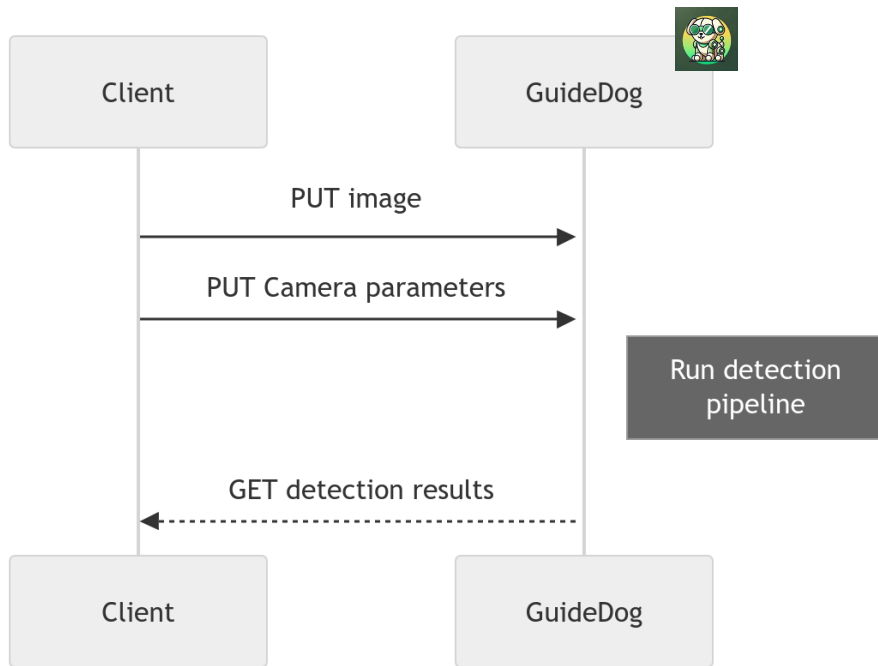
Upload image



PCVP Detection

Detect!

Current API definition



Pipeline Setup

- **Model-based 6D pose estimation**
- **Agree on limited set of objects beforehand**
- **Fine-tune on real data**

Open Questions



How to provide access?

- Webserver
 - More preparation time from DLR, but hopefully accessible before September
 - Internet access is not guaranteed in EU parliament
- Offline Computer in local network
 - Setup during the preparation week in Brussels

Performance?

- Speed: We need to profile inference and network delays
- Precision: Collect data to fine-tune weights